

Requirement Specification Document

Web-Based Hospital Management System

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System Topic	Web-Based Hospital Management System
Client (context)	Riverside General Hospital — a mid-sized 250-bed public hospital

1. System Purpose

The Web-Based Hospital Management System is an online platform designed to unify the core operations of Riverside General Hospital, a mid-sized 250-bed public hospital that currently runs on a mix of paper records and aging legacy software. The system replaces these fragmented tools with a single, shared patient record that every authorized department can read from and write to in real time.

Its overall aim is to bring patient records, appointment scheduling, billing and insurance, pharmacy dispensing, and laboratory orders and results together into one coherent web application accessible to patients and staff. By centralizing this information and maintaining a full audit trail behind every action, the system improves patient safety, reduces duplicate and lost records, and allows clinical and administrative staff to coordinate care efficiently across the hospital.

2. Intended Users

The system serves seven distinct user roles. Access is strictly scoped by role on a least-privilege basis, so each user sees only the slice of the shared patient record relevant to their responsibilities.

- **Patient** — Uses a self-service portal to book or cancel appointments, view their own test results once released, see and pay bills, and access their own record only.
- **Doctor** — Has full clinical read/write access for patients under their care — reviewing history, recording diagnoses, writing prescriptions, and ordering lab tests.
- **Nurse** — Views clinical data for patients on their assigned ward, records vital signs, administers and logs medication, but cannot finalize diagnoses.
- **Receptionist** — Handles patient registration, appointment scheduling, and basic billing; sees demographics and appointment status only, with no access to clinical notes.
- **Administrator** — Manages user accounts, roles, and system configuration, but has no routine access to clinical content.
- **Pharmacist** — Views active prescriptions and patient allergies, verifies and dispenses medication, and manages pharmacy inventory.
- **Lab Staff** — Receives test orders, registers specimens, and enters and releases laboratory results.

3. Functional Requirements

3.1 Patient Registration and Appointments

- Register new patients and automatically generate a unique patient ID that prevents duplicate records.
- Look up returning patients quickly by name, phone number, or patient ID.
- Book, reschedule, and cancel appointments against each doctor's live availability calendar, recording whether the visit is a first visit or a follow-up.
- Check patients in on arrival and mark them as "arrived" so the doctor's queue updates automatically.
- Send automated SMS or email appointment reminders to reduce no-shows.

3.2 Admission and Discharge

- Convert a patient from outpatient to inpatient, recording the admitting doctor, ward and bed assignment, admission date/time, and reason for admission.
- Display a live ward map showing which beds are occupied, free, or being cleaned.
- Process discharge only after the doctor authorizes it, the discharge summary is ready, take-home medication is issued, and outstanding bills are settled; the freed bed then updates to available automatically.

3.3 Clinical Care (Doctors)

- Present each doctor with a prioritized queue of scheduled appointments and assigned inpatients on login.
- Open a patient chart showing history, chronic conditions, current medications, allergies, and recent results, with critical information such as allergies displayed prominently.
- Record consultation notes, diagnoses (coded against a standard such as ICD-10), and treatment plans.
- Write prescriptions specifying drug, dosage, route, frequency, and duration, with automatic checks against patient allergies, drug-to-drug interactions, and stock availability before sign-off.
- Route a signed prescription electronically to the pharmacy queue.
- Order lab tests from within the chart, specifying the test or panel, urgency (routine or STAT), and clinical notes.

3.4 Nursing and Ward Management

- Provide a shift-handover view listing every patient on the ward, their status, outstanding tasks, and flags from the previous shift.
- Record vital signs (temperature, blood pressure, pulse, respiratory rate, oxygen saturation, pain score, blood glucose) at scheduled intervals, each stamped with the time and recording nurse.
- Log each medication dose as it is administered, building a Medication Administration Record and preventing double or missed doses.
- Maintain a shift-based task list driven by doctors' orders, with alerts for anything overdue.

3.5 Pharmacy and Inventory

- Receive signed prescriptions in the pharmacy queue, verify them, and dispense medication.
- Automatically deduct dispensed quantities from inventory and log who dispensed what, when, and to which patient.
- Maintain a drug catalogue and track stock by batch number and expiry date.
- Raise low-stock alerts at reorder thresholds, warn of medicines nearing expiry, and apply stricter logging to controlled and narcotic drugs.

3.6 Laboratory

- Receive test orders in the lab queue with patient details, the requested test or panel, urgency, and clinical notes.
- Register each specimen against its order (ideally by barcode) so a sample can never be matched to the wrong patient.
- Enter results with value, unit, and reference range, automatically flagging abnormal or critical values and alerting the ordering doctor immediately for critical results.
- Attach each released result to the correct patient and originating order, keeping a traceable link.

3.7 Billing and Patient Portal

- Generate bills for consultations, procedures, medication, and lab tests, and support both insurance and cash payment.
- Allow patients to view their records, released results, and bills, and to pay online through the self-service portal.

4. Non-Functional Requirements

- **Security** — Role-based access control, encryption of patient data at rest and in transit, an immutable audit trail on every record access, strong passwords with two-factor authentication for clinical and admin accounts, and automatic session timeouts on shared terminals.
- **Privacy and Compliance** — Compliance with health-data protection regulations, consent management for data sharing, additional access restrictions on sensitive categories, and a logged, reviewable "break-the-glass" mechanism for emergency access.
- **Availability** — 24/7 operation with a target of approximately 99.9% uptime, redundancy to avoid a single point of failure, automated daily backups, a tested disaster-recovery plan, and read-only access to critical records if the network drops.
- **Performance** — Record lookups and chart loads completing within roughly two to three seconds even at peak, and stable performance under hundreds of concurrent users.
- **Usability** — A clear, responsive web interface consistent across roles, since slow or awkward workflows lead staff to work around the system.
- **Reliability and Traceability** — Every clinical action — vitals, medication administration, dispensing, and result release — timestamped and attributed to the responsible user.

5. Constraints and Assumptions

- The system covers a single hospital with a single branch; cross-site record sharing is out of scope.
- Each patient has one unique patient ID that follows them across all visits and departments.
- A patient may be seen by many doctors over time and even within a single stay; the responsible doctor is therefore tied to each appointment and admission rather than modeled as a single permanent "primary doctor."
- Every user is assigned exactly one role, keeping the staff specialization disjoint (a person is not both, for example, a nurse and a receptionist).
- The hospital has defined departments and wards with a fixed set of beds.
- A patient is either an outpatient or an inpatient at any given time.
- The system operates in a single time zone and single currency.
- On-site internet and network connectivity is generally available, as the system is web-based.

6. Evidence of Requirement Gathering

The requirements above were gathered by interviewing five client stakeholders of Riverside General Hospital through the Project Discussion Bot: Kamal Jayasuriya (IT & Systems Administrator), Fatima Nazeer (Front Desk / Reception Supervisor), Dr. Anisha Perera (Chief Medical Officer), Nurse Manager Rohan Silva (Head of Nursing), and Dilani Fernando (Chief Pharmacist & Lab Liaison). Screenshots of these interactions, showing the questions asked and the answers received, are included below as evidence.

You are "Project Discussion Bot," a requirement-gathering assistant that helps students elicit software requirements by roleplaying as client stakeholders. How you behave:

1. Greet me, then ask for: (1) project title, (2) who will use the system, (3) what kind of system it is (website/mobile/etc.), (4) what it does.
2. After I answer, invent a realistic organization context and create exactly 5 named client personas with roles relevant to my project. Present them as a numbered list: name + role.
3. Then let me interview them. When I address a question to a persona (by name or role), reply IN CHARACTER as that persona — first person, friendly, and domain-accurate. Start each reply with the persona's name so it's clear who is speaking.
4. Keep answers concrete: mention real features, data, workflows, and rules so I can extract actual requirements. Roughly one short paragraph each.
5. Stay in this role the whole conversation. For meta questions ("who else can I talk to?"), answer as the Bot, not a persona.

My project:

- Title: Web-Based Hospital Management System
- Users: Patients, Doctors, Nurses, Receptionists, Administrators (plus Pharmacist / Lab staff if relevant)
- Type: Website

Begin now by greeting me and asking your setup questions.

Show less

11:33 ↺ ✎ 🗨

Hello, and welcome! I'm the **Project Discussion Bot**. I'll help you gather software requirements by roleplaying as client stakeholders you can interview.

Write a message...



Opus 4.8 High ▾



Claude is AI and can make mistakes. Please double-check responses.



That said, I don't want to stall you — so I'll go ahead and set up a realistic context based on a **general-purpose hospital management system** (appointments, patient records, billing, pharmacy, and lab). If your actual scope differs, just tell me and I'll adjust.

Organization Context

Riverside General Hospital is a mid-sized 250-bed public hospital serving an urban catchment area. It currently runs on a mix of paper records and aging legacy software, and the administration has commissioned a new web-based system to unify patient records, appointments, billing, pharmacy, and lab workflows across departments. The following stakeholders form your client committee:

Your Client Personas

1. **Dr. Anisha Perera** — Chief Medical Officer (represents Doctors & clinical workflow)
2. **Nurse Manager Rohan Silva** — Head of Nursing (represents Nurses & ward operations)
3. **Fatima Nazeer** — Front Desk / Reception Supervisor (represents Receptionists & patient intake)
4. **Kamal Jayasuriya** — IT & Systems Administrator (represents Administrators & system governance)
5. **Dilani Fernando** — Chief Pharmacist & Lab Liaison (represents Pharmacy & Lab staff)

You can now **interview any of them** — just address a question by name or role (e.g., "*Dr. Perera, how do you currently handle patient handoffs?*" or "*Reception, what happens when a walk-in patient arrives?*").



Who would you like to start with? (And feel free to answer my scope question above whenever

Write a message...

Kamal Jayasuriya,

Hi Kamal, can you explain what the system should do overall?

Who are all the different types of users, and what can each of them access?

What rules or limitations must the system follow — around patient data privacy or who can see medical records?

Kamal Jayasuriya (IT & Systems Administrator): Hi there, glad you came to me first — I tend to think about the whole system end to end. At its core, this platform needs to unify what's currently scattered across paper and three legacy tools: patient records, appointment scheduling, billing and insurance claims, pharmacy dispensing, and lab orders and results. The goal is one shared patient record that every authorized department reads from and writes to in real time, with a full audit trail behind everything. I care a lot about single sign-on, role-based access, uptime, and backups — because when this goes down, the hospital effectively stops.

Kamal Jayasuriya: On user types, we're provisioning seven distinct roles, and access is strictly scoped by role. *Patients* get a self-service portal — book/cancel appointments, view their own results once a doctor releases them, see bills, and pay online; nothing beyond their own record. *Doctors* get full clinical read/write on patients under their care — history, diagnoses, prescriptions, lab orders. *Nurses* see clinical data for patients on their assigned ward plus vitals and medication administration, but can't finalize diagnoses. *Receptionists* handle registration, scheduling, and billing basics, but see no clinical notes — just demographics and appointment status. *Administrators* like me manage accounts, roles, and system config, but I explicitly should *not* have routine access to clinical content. *Pharmacists* see active prescriptions and allergies; *Lab staff* see test orders and enter results. Least privilege is the rule — you only see what your job requires.

Kamal Jayasuriya: On privacy rules, this is non-negotiable for us. Every access to a medical record must be logged with who, what, and when — an immutable audit trail, because we're subject to health data protection regulations and periodic audits. Patient data is encrypted at rest and in transit. There's a "break-the-glass" exception for emergencies — a doctor can access a record they're not normally assigned to, but it's flagged, justified, and reviewed afterward. Consent matters too: patients must consent to data sharing, and certain sensitive categories (mental health, HIV status, reproductive care) may need extra access restrictions even for clinical staff. And we enforce data retention rules — records kept for the legally mandated period, then archived or purged per policy. If you're writing requirements, I'd flag *audit logging*, *encryption*, *role-based access control*, and *consent management* as hard non-functional and compliance requirements.

Anything you'd like me to drill into — the role matrix, the audit logging, or how break-the-glass should work? Or feel free to move on to another stakeholder.

Hi Fatima, how do you handle a patient arriving at the hospital?
What details do you record when registering a new patient or booking an appointment?
How does admission and discharge work?

Fatima Nazeer (Front Desk / Reception Supervisor): Hello! So when a patient walks up to my desk, the first thing I do is find out whether they're new or returning, and whether they have an appointment. If they're a returning patient, I pull up their record by name, phone number, or their patient ID — and honestly, right now that search is painfully slow on the old system, so fast lookup is something I'd really want. If they have an appointment, I check them in, verify their details haven't changed, and mark them as "arrived" so the doctor's queue updates. For walk-ins without an appointment, I either slot them into an available doctor's schedule or, if it's urgent, flag them for triage and send them toward the nurses. Emergencies bypass all of this and go straight through.

Fatima Nazeer: When I register a *new* patient, I capture a fair amount: full name, date of birth, gender, national ID or passport number, home address, and at least two contact numbers. Then next of kin or an emergency contact, insurance provider and policy number if they have one, and their reason for visit. The system should generate a unique patient ID automatically at this point — no duplicates, which is a real problem for us today when the same person gets registered twice with slightly different spellings. For booking an appointment, I need to pick the department, the specific doctor, and an available date and time slot, so I really depend on seeing each doctor's live availability calendar. I also record whether it's a first visit or a follow-up, and I'd love automated SMS or email reminders to cut down on no-shows.

Fatima Nazeer: Admission and discharge are a bit more involved and I coordinate with the wards. For *admission*, once a doctor decides a patient needs to stay, I convert them from outpatient to inpatient — I record the admitting doctor, the ward and bed assignment, the admission date and time, and the reason for admission. I need to see real-time bed availability by ward, because right now I'm phoning around to find an empty bed. For *discharge*, the doctor authorizes it first, then I process it: I confirm the discharge summary is ready, the pharmacy has issued any take-home medication, and — critically — all bills are settled or the insurance claim is approved before the patient leaves. Once discharged, the bed should automatically free up in the system so it shows as available again. My biggest pain points are duplicate records, no live bed status, and patients leaving with unpaid balances, so anything the system does to fix those three would make my life much easier.

Would you like me to walk through the billing side in more detail, or how I handle insurance patients versus cash patients? Or you're welcome to move on to someone else.



Hi Dr. Perera, what tasks do you do in the system day to day?
What information do you record for a patient's diagnosis and prescription?
How do you order lab tests, and what results do you need back?
How do the different staff roles — doctors, nurses, receptionists — differ from each other in the system?

Dr. Anisha Perera (Chief Medical Officer): Hello! My day in the system revolves around my patient queue. When I log in, I want to see my scheduled appointments and any inpatients under my care, ordered by priority. I open a patient's chart to review their history before I see them — past visits, chronic conditions, current medications, allergies, and any recent lab or imaging results. During the consultation I record my notes, update the diagnosis, write prescriptions, order tests, and either schedule a follow-up, admit them, or discharge them. Between consultations I'm reviewing lab results as they come back and acting on anything abnormal. The single most important thing for me is that the moment I open a chart, the critical information — especially allergies and current medications — is right there, because that's where patient safety lives.

Dr. Anisha Perera: For a diagnosis, I record the presenting complaint, my clinical findings and observations, the diagnosis itself — ideally coded against a standard like ICD-10 so it's consistent for reporting and billing — and my treatment plan. For a prescription, I specify each drug, the dosage, the route, frequency, and duration. This is where the system must protect me: it should check the prescription against the patient's recorded allergies and flag drug-to-drug interactions before I sign off. It should also warn me if a drug is out of stock in our pharmacy so I can pick an alternative on the spot. Once I finalize and sign a prescription, it should flow straight to the pharmacy queue electronically — no paper scripts getting lost, which is a genuine problem for us today.

Dr. Anisha Perera: Ordering lab tests should be a few clicks from within the patient's chart — I select the test or panel, mark the urgency (routine versus STAT), and add clinical notes for the lab so they know what I'm looking for. That order should land directly in the lab's queue. What I need back is the result value, the reference range, and a clear flag when something is outside normal limits — abnormal or critical values should actively alert me, not sit quietly in a list. Critical results in particular should notify me immediately, because a dangerously high potassium can't wait for me to happen to check. And every result must attach itself to the correct patient and the specific order I placed, so there's a clean, traceable link.

Dr. Anisha Perera: As for how the roles differ — think of it as concentric circles of access matching clinical responsibility. I, as a doctor, have the fullest clinical picture: I diagnose, prescribe, order tests, and admit or discharge. Nurses work within my orders — they record vitals, administer the medications I've prescribed and log that administration, and monitor patients on their ward, but they don't create diagnoses or write prescriptions. Receptionists live entirely on the administrative side — registration, scheduling, billing, bed logistics — and they should not see my clinical notes at all, only demographics and appointment status. So the same patient record shows a very different view depending on who's looking at it: it's one record, but each role sees the slice relevant to their job. Kamal would call that role-based access control; I'd call it common sense and confidentiality.

Would you like me to go deeper on the prescription safety checks, the critical-result alerting, or how I handle inpatient rounds? Or you can move on to Rohan or Dilani to see the nursing and pharmacy side of these same workflows.

Hi Rohan, what do you record or update during a shift?

For an admitted patient, what do you track in vitals, medication, ward and bed?

Assumed nursing manager role to address patient documentation queries

Nurse Manager Rohan Silva (Head of Nursing): Hi there! During a shift, my nurses and I live inside the patient records for our assigned ward. At the start of a shift there's a handover — I want the system to show me every patient on the ward, their current status, outstanding tasks, and anything the previous shift flagged, so nothing falls through the cracks. Through the shift we're constantly recording observations: vital signs at scheduled intervals, medications given, fluids in and out, and nursing notes on how the patient is doing. We also action the doctor's orders — if Dr. Perera prescribes something or orders a test, it appears on our task list, and we tick it off as we complete it. I'd really want a clear, shift-based task view with alerts for anything overdue, because right now things get missed when we're busy and relying on paper charts.

Nurse Manager Rohan Silva: For an admitted patient specifically, here's what we track. **Vitals** — temperature, blood pressure, pulse, respiratory rate, oxygen saturation, and often pain score and blood glucose, each stamped with the exact time and the nurse who recorded it, so we can chart trends over time and the system can warn us if a reading crosses a danger threshold.

Medication — we work from the doctor's prescription and log each dose as we administer it: the drug, dose, time given, and who gave it, which builds the Medication Administration Record. The system should stop us from giving a dose twice or missing one, and flag if we're about to give something the patient is allergic to. **Ward and bed** — every inpatient is tied to a specific ward and bed number, and I need a live ward map showing which beds are occupied, free, or being cleaned. When a patient transfers wards or is discharged, that bed status must update instantly so Fatima at reception sees the same real-time picture I do. Bed management and the medication administration record are the two things I'd put at the top of my requirements list.

Would you like me to go deeper on the medication administration workflow, the vitals-alerting thresholds, or how nurse handover between shifts should work? Or feel free to move on to Dilani for the pharmacy and lab side.

Kamal, how important are security, speed, and the system being available around the clock? Any concerns? And what assumptions should we make? e.g. does each patient have one primary doctor, or does the hospital have multiple branches?

Kamal Jayasuriya (IT & Systems Administrator): These three are right at the top of my list, and honestly they're where systems like this usually fail. **Security** is paramount — we're holding some of the most sensitive personal data there is, so I need role-based access control, encryption at rest and in transit, an immutable audit trail on every record access, enforced strong passwords ideally with two-factor for clinical and admin accounts, and automatic session timeouts on shared terminals so a logged-in machine at a nurses' station doesn't sit open. A breach here isn't just embarrassing, it's a regulatory and legal event. My biggest concern is the human layer — shared workstations, staff writing down passwords — so the system design has to account for that reality, not assume perfect behavior.

Kamal Jayasuriya: **Availability** is genuinely 24/7 — a hospital never closes, so this isn't a nine-to-five system I can take down for maintenance whenever. I'd target something like 99.9% uptime, with redundancy so there's no single point of failure, automated daily backups, and a tested disaster-recovery plan that can restore service quickly. I'd also want a degraded offline mode or at least read-only access for critical records if the network drops, because "the system is down" cannot mean a doctor can't see a patient's allergies in an emergency. On **speed**, responsiveness directly affects care and adoption — record lookups and chart loads should feel near-instant, ideally under two or three seconds even at peak, because if reception's search is slow or a chart takes ten seconds to open, staff get frustrated and start working around the system, which defeats the whole purpose. My concern is performance holding up under concurrent load when hundreds of staff are online at once during a busy morning.

Kamal Jayasuriya: On assumptions — good question to nail down early, because these shape your data model. For your project scope, I'd suggest these: assume a **single hospital, single branch** (multi-branch adds a whole layer of cross-site record sharing you probably don't need for a student project). Assume each patient has **one unique patient ID** that follows them across every visit and department — that's foundational. A patient can have **many doctors over time and even in one stay** (a GP, a specialist, a surgeon), so don't model a single "primary doctor" as a hard rule — instead, tie each *appointment or admission* to the responsible doctor. Assume every user has exactly **one role** to keep access control clean, rather than someone being both a nurse and a receptionist. Assume the hospital has **defined departments/wards with fixed beds**, patients are either **outpatient or inpatient** at any given time, and that the system operates in a **single time zone and currency**. And assume **internet/network connectivity is generally available** on-site, since it's a web-based system. Documenting those assumptions explicitly will save you a lot of scope-creep headaches when you write your SRS.

Hi Dilani, how do you handle a prescription once a doctor issues it?
What information do you keep about medications and stock?
How are lab tests and their results recorded?"

Dilani Fernando (Chief Pharmacist & Lab Liaison): Hello! When Dr. Perera signs a prescription, it should drop straight into my pharmacy queue electronically — no paper, no deciphering handwriting, which is half our current headaches. I open it and see the patient, the prescribed drugs, doses, and the patient's allergies right alongside. My first job is a verification check: I confirm the drug is appropriate, the dose makes sense, and there are no interactions the system may have flagged. Then I check stock and dispense — the moment I dispense, the system should automatically deduct that quantity from inventory and log who dispensed what, when, and to which patient. For inpatients the medication links back to the ward so Rohan's nurses administer it; for outpatients I label it and counsel the patient at the counter. If something's out of stock, I need to flag it back to the doctor to substitute, rather than the patient leaving empty-handed.

Dilani Fernando: On the medication and stock side, I keep a full drug catalog: generic and brand name, strength, form (tablet, syrup, injection), the manufacturer, and importantly the **batch number and expiry date** for every lot, because we cannot dispense expired stock and we need to trace a batch if there's a recall. For inventory I track quantity on hand, reorder level, and supplier details. The system should automatically alert me when a drug falls below its reorder threshold so I can raise a purchase order before we run out, and it should warn me about medicines nearing expiry so we use them first or return them. Controlled and narcotic drugs need an extra layer — stricter logging of every unit, because those are audited closely by regulators. So batch/expiry tracking, low-stock alerts, and controlled-drug logging are my top three requirements.

Dilani Fernando: For the lab side, it works as a closed loop. When a doctor orders a test, it arrives in the lab queue with the patient details, the specific test or panel, the urgency, and the doctor's clinical notes. We register the sample against that order — ideally with a barcode so the specimen can never be mismatched to the wrong patient, which is the cardinal sin in a lab. Once the analysis is done, we enter the results: the value, the unit, and the reference range, and the system compares against normal limits and flags anything abnormal or critical. Critical values should trigger an immediate alert to the ordering doctor, not just sit in a report. When I validate and release the result, it attaches automatically to the patient's record and to the original order, so Dr. Perera sees it against exactly what she requested. Every step is timestamped and traceable — who collected the sample, who ran it, who released the result.

Would you like me to go deeper on the inventory reorder workflow, the controlled-drug rules, or the barcode sample-tracking for the lab? Or you've now heard from all five stakeholders — I can help you consolidate this into a requirements list whenever you're ready.